

TrES-3b

R-0521

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240913 · created 2026-07-21T00:32:53+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460566.6341 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.039
Transit depth (Rp/Rs)^2	3.43 +/- 1.46 [%]
Orbital Inclination (inc)	81.88 +/- 1.62
Airmass coefficient 1 (a1)	1.02 +/- 0.014
Airmass coefficient 2 (a2)	-0.017 +/- 0.055
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	None
Optimal Aperture	3.66
Optimal Annulus	7.01
Transit Duration (day)	0.0282 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.185 ± 0.039 vs 0.16309 (deviation 0.52σ)
Duration fit vs published	0.0282 d vs 0.0567 d (deviation 2.75σ)
Mid-time O–C	3.4 min
Depth SNR	4.4
Residual scatter	1.36 %

Light curve

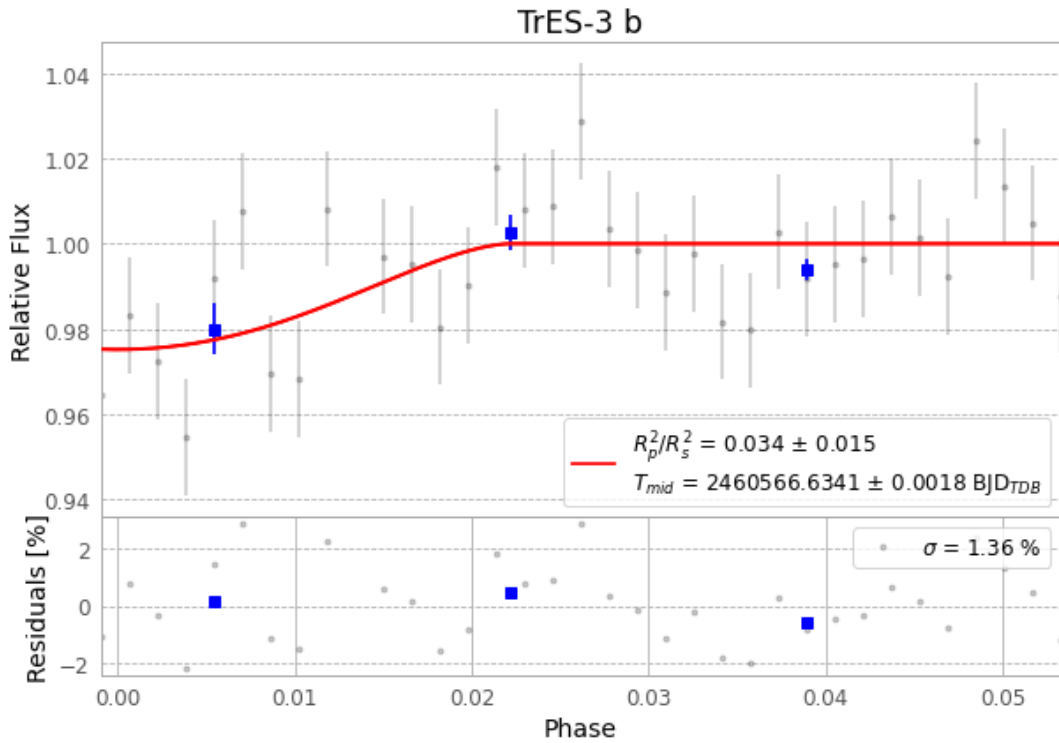


Figure 1 — FinalLightCurve_TrES-3 b_2024-09-12.png (SHA-256 bec30b217e5a0daf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [332, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3aa1bb16b9bff7d5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 4685edf

Data provenance

35 FITS frames (23 MB), TRES-3240913030915.FITS → TRES-3240913045115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240913.log (e5d259356c7a...), FinalLightCurve_TrES-3 b_2024-09-12.png (bec30b217e5a...), FinalLightCurve_TrES-3 b_2024-09-12.csv (2b7fb6534d98...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 00:32Z.